DO NOW — RECALL

Processors, scheduling and LMC practice

Name: _____ Class: _____ Date: _____

1. What does LMC STA do?

2. What is the role of MAR?

3. What is a time quantum?

4. Full adder 1+1+0: sum and carry bits?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Copies ACC contents to the specified mailbox.

2. Model answer

Address of the memory location being accessed.

3. Model answer

Maximum CPU slice before RR rescheduling.

4. Model answer

10_2: sum0, carry1.

By the end, you can…

• Trace the LMC accumulator and mailbox separately: INP overwrites ACC, while STA copies to storage.

• Schedule independent CPU jobs with an explicit arrival/time rule and label every Gantt boundary.

• Distinguish processor speed, memory cache and interrupt/OS services when explaining performance.

Vocabulary — know these terms

ACC — Accumulator register.

FCFS — First Come First Served.

SJF — Shortest Job First.

pipeline — Overlap instruction stages.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain and check

Explain the reason for the result.

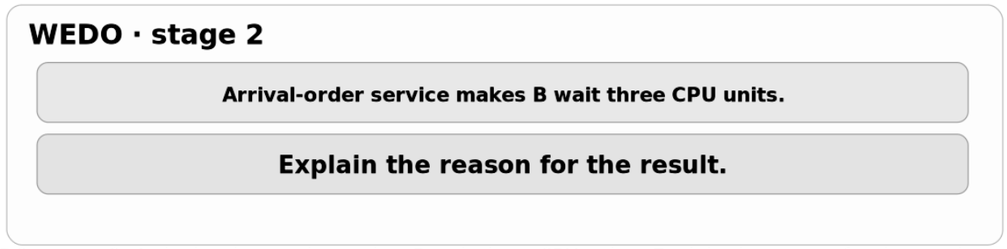
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

FCFS all tasks t0: A=3,B=2. Show timeline and mean waiting time.

Arrival-order service makes B wait three CPU units.



WEDO · stage 2: Arrival-order service makes B wait three CPU units.; Explain the reason for the result.

NEW LEARNING

Read, annotate and explain

Understand

Trace the LMC accumulator and mailbox separately: INP overwrites ACC, while STA copies to storage.

Apply

Schedule independent CPU jobs with an explicit arrival/time rule and label every Gantt boundary.

Evaluate

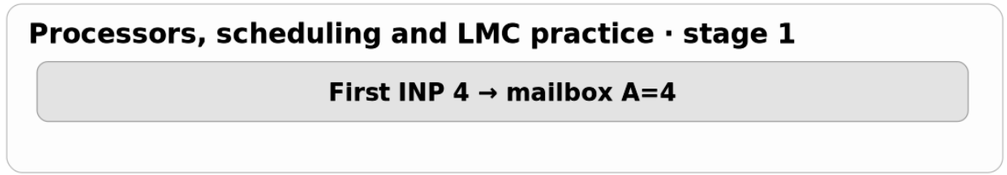
Distinguish processor speed, memory cache and interrupt/OS services when explaining performance.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

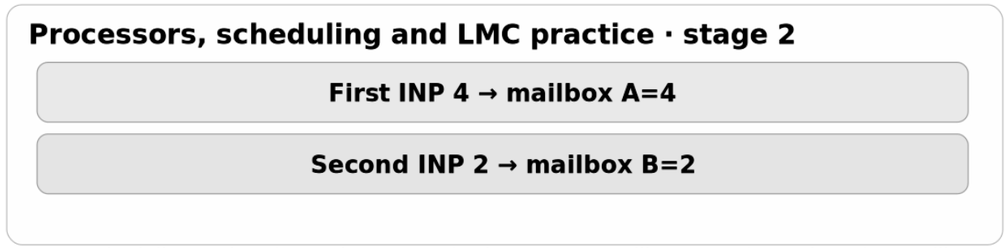
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



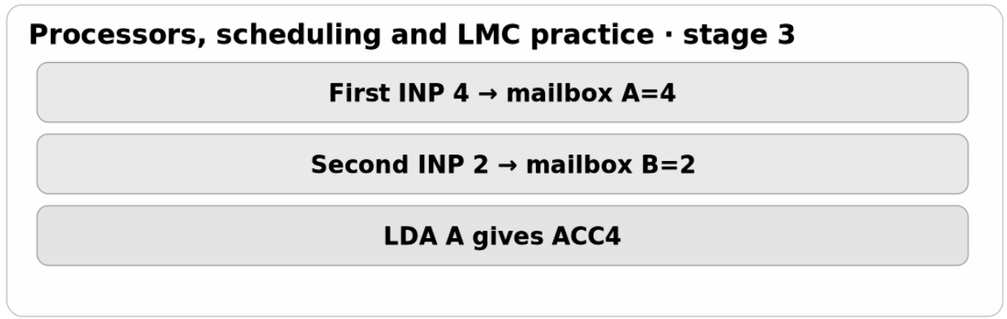
Processors, scheduling and LMC practice · stage 1: First INP 4 → mailbox A=4

New Learning



Processors, scheduling and LMC practice · stage 2: First INP 4 → mailbox A=4; Second INP 2 → mailbox B=2

New Learning



Processors, scheduling and LMC practice · stage 3: First INP 4 → mailbox A=4; Second INP 2 → mailbox B=2; LDA A gives ACC4

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

LMC STA X loads ACC from mailbox X.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

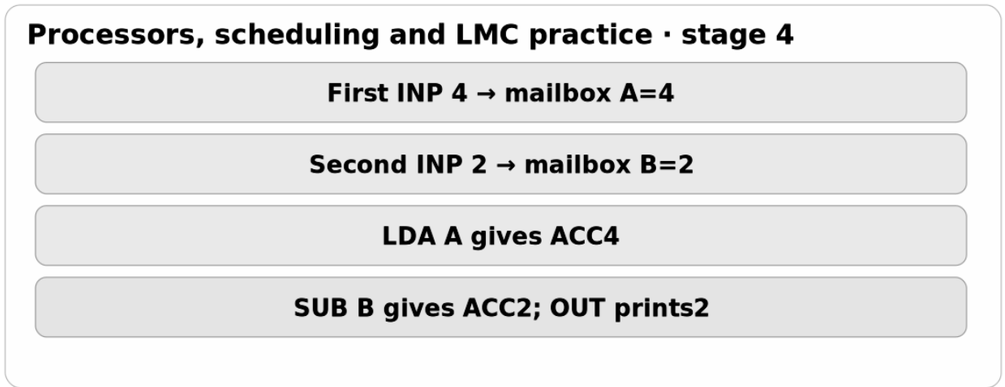
Question 3

Evaluate clock speed, cache and pipelining for a task with heavy memory latency and branches. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Processors, scheduling and LMC practice · stage 4: First INP 4 → mailbox A=4; Second INP 2 → mailbox B=2; LDA A gives ACC4; SUB B gives ACC2; OUT prints2

I DO — WORKED EXAMPLE

Follow the worked method

Trace sequential instruction fetch when PC=90.

Predict first

PC holds address of next instruction, not the instruction contents.

Explain and check

Explain the reason for the result.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Using LMC code above, trace ACC after each input, LDA, SUB and OUT. [6]

Question 2

Processes A (burst 4), B (burst 2) and C (burst 1) all arrive at t=0. For FCFS, use the stated arrival-list order A, B, C to break ties; for non-preemptive SJF choose the shortest available job. Compare execution orders and mean waiting times. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Arrival-order service makes B wait three CPU units.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace sequential instruction fetch when PC=90.

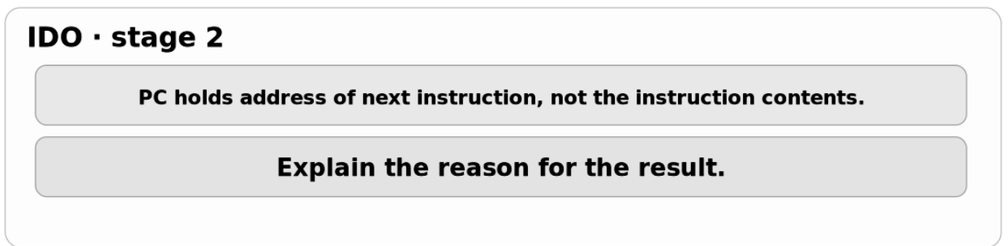
PC holds address of next instruction, not the instruction contents.



IDO · stage 1: PC holds address of next instruction, not the instruction contents.

Trace sequential instruction fetch when PC=90.

PC holds address of next instruction, not the instruction contents.



IDO · stage 2: PC holds address of next instruction, not the instruction contents.; Explain the reason for the result.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

FCFS all tasks t0: A=3,B=2. Show timeline and mean waiting time.

Arrival-order service makes B wait three CPU units.

WEDO · stage 1: Arrival-order service makes B wait three CPU units.